

Memory-Aware Feature Design Dominates Quantum-Inspired Architecture in Physics-Informed Solvers for Time-Fractional Integro-Differential Equations

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Abstract—Physics-informed neural networks for time-fractional integro-differential equations must approximate a nonlocal Caputo derivative together with a weakly singular Volterra memory term, and their accuracy depends both on the numerical treatment of those operators and on the representation given to the network. Architecture-level proposals, including quantum-inspired trainable-embedding designs, and feature-level priors that encode fractional-memory structure are frequently introduced together, which makes it difficult to attribute any observed gain to one or the other. This letter separates them. We benchmark a classical solver, a quantum-inspired trainable-embedding surrogate, and analytic memory-aware input features on a manufactured fractional integro-differential problem under a fairness-controlled protocol with matched parameter budgets, matched collocation budgets, and repeated seeds, and we add a control whose only difference from the classical baseline is the input representation. Analytic memory features emerge as the dominant effect, improving accuracy by substantially more than any architectural change we tested, and the improvement holds on both the classical and the quantum-inspired track. The trainable-embedding surrogate contributes a smaller but independent gain under matched inputs, and the combined model is the strongest variant, although its advantage over the representation-matched control is not established at the primary operating point. A normalization variant intended to stabilize the surrogate is found to be indistinguishable from the surrogate itself. Conclusions are reported against an explicit zero-predictor reference and are shown to depend on training to convergence. No claim of quantum advantage is made: the surrogate is classical, and its sinusoidal feature map has learned, input-dependent frequencies.

Index Terms—Physics-informed neural networks, fractional integro-differential equations, Caputo derivative, product integration, graded meshes, quantum-inspired neural networks, memory-aware features.

I. INTRODUCTION

TIME-FRACTIONAL integro-differential systems are a demanding test for neural solvers. Their operators are nonlocal in time, so the state at t depends on accumulated history rather than on local derivatives alone, and the kernels involved are weakly singular. Automatic differentiation cannot express a Caputo derivative directly, so physics-informed neural networks (PINNs) for such problems must embed a numerical history operator inside the residual [1], [2].

Two directions have been proposed for improving on a baseline PINN in this setting. The first is architecture-level: trainable-embedding quantum physics-informed neural networks (TE-QPINNs) are a recent instance, in which a learned embedding drives an angle-encoded feature map [3], [4]. The second is feature-level: analytic priors that place fractional-memory structure directly in the network input. These are frequently introduced together, and the consequence is an attribution problem. A quantum-inspired architecture evaluated against a plain baseline can appear to succeed when the gain in fact comes from an accompanying change of input representation, and nothing in such a comparison distinguishes the two explanations.

PINNs were introduced for forward and inverse problems governed by nonlinear PDEs [1], and were extended to fractional operators by replacing the inexpressible derivative with a numerical discretization inside the residual [2]. That substitution is what makes the present setting distinctive: the residual is no longer local, so the cost and the accuracy of the solver are both governed by how the history is treated. The L1 scheme on a graded mesh is the standard device for the Caputo term when the solution has a weak singularity at the origin, and the grading exponent controls the attainable order [5], [6]. For the weakly singular Volterra term, product integration handles the kernel analytically instead of resolving it by quadrature [7], [8]. On the architecture side, hybrid quantum-classical models parameterize a circuit or an emulation of one [3], and trainable-embedding variants learn the encoding rather than fixing it [4]. Sinusoidal encodings of this kind are closely related to Fourier-feature models in classical deep learning [9], a connection we return to below.

This letter separates them. Four variants are evaluated against a classical baseline under one residual construction, matched parameter counts, matched collocation budgets, and ten seeds. The decisive addition is a *classical-plus-memory* control whose only difference from the baseline is the input representation: the architecture, the residual, the sampling, and the loss weights are unchanged. Any accuracy this control recovers is attributable to representation alone, and whatever the quantum-inspired surrogate achieves beyond it is attributable to architecture. This control is absent from most quantum-inspired PINN comparisons, and it is what makes the attribution possible.

The contributions are a fairness-controlled benchmark that

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separates feature-level from architecture-level gains on a fractional integro-differential problem; an attribution of most of the available improvement to the input representation rather than to the architecture; a smaller but independent architectural gain under matched inputs; a negative result for the normalization variant intended to stabilize the surrogate; and a protocol argument that comparisons of this kind should carry both a representation-matched control and an explicit zero-predictor reference.

II. PROBLEM AND RESIDUAL CONSTRUCTION

A. Governing Equation

We solve the time-fractional integro-differential equation

$$D_t^\alpha u(x, t) - (x^2 + 1)u_{xx}(x, t) + \int_0^t \sin(x)(t-s)^{-\beta} u(x, s) ds = f(x, t) \quad (1)$$

on $x \in [0, 1]$, $t \in (0, 1]$, with $u(0, t) = t^\alpha$, $u(1, t) = -t^\alpha$ and $u(x, 0) = 0$. Here D_t^α is the Caputo derivative

$$D_t^\alpha u(x, t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} \partial_s u(x, s) ds, \quad (2)$$

and the Volterra kernel $(t-s)^{-\beta}$ is weakly singular but integrable for $0 < \beta < 1$. The manufactured solution is $u_{\text{exact}}(x, t) = t^\alpha \cos(\pi x)$, with f derived analytically under the full operator, so that final L_2 and L_∞ errors are computable exactly and identically across variants.

B. Discretization

The Caputo term uses an L1 approximation on a graded temporal mesh $t_n = T(n/N)^r$ with $r \geq 1$ [5], [6], which concentrates nodes near $t = 0$ where the temporal derivative of the solution is singular. The Volterra term uses kernel-aware product integration: on each interval $[t_j, t_{j+1}]$ the solution is interpolated linearly while the singular kernel moments $\int \tau^{-\beta} d\tau$ and $\int \tau^{1-\beta} d\tau$ are integrated analytically [7], [8], giving

$$\mathcal{I}_n[u](x) \approx \sum_{j=0}^n w_{n,j} \hat{u}(x, t_j), \quad (3)$$

with weights $w_{n,j}$ depending only on the mesh and β . Verified against the exact Beta-function value for $u = s^\alpha$, this rule converges at second order, whereas applying Gauss–Legendre quadrature to the singular integrand stagnates near 10^{-2} regardless of mesh refinement and is therefore unusable as a reference.

Both the L1 stencil and (3) are linear in the history values $u(x, t_j)$. Each is therefore a fixed weight matrix applied to one history tensor, which we assemble with a single batched network evaluation rather than one evaluation per history node. The rearrangement is algebraically exact: residuals and parameter gradients agree with the naive loop to floating-point precision. It reduces residual cost by about two orders of magnitude, which is what makes a ten-seed protocol at a converged budget practical on modest hardware.

The network minimizes $\mathcal{L} = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}$ with weights held fixed across all variants.

III. MODEL VARIANTS

A. Trainable-Embedding Surrogate

The surrogate is a classical model that emulates TE-QPINN design ideas [4] without a quantum circuit. Inputs are affine-rescaled; a trainable embedding network produces ϕ_i , from which angle-style coordinates $\theta_i = \phi_i(x, t) \tilde{z}_i$ are formed against cycled input coordinates. The feature map is $[\sin \theta, \cos \theta]$, augmented with pairwise terms $\sin(\theta_i) \cos(\theta_{i+1})$, followed by a mixing block and an expectation-style readout with an optional residual branch.

We state plainly what this is: a multilayer perceptron on a sinusoidal feature map whose frequencies are learned and input-dependent, that is, a learned Fourier-feature model [9]. The pairwise terms are products of independently computed scalars and therefore factorize; they are degree-two interaction features, not entanglement. No quantum hardware or simulator is used, and no quantum advantage is claimed.

B. Analytic Memory Features

The memory-aware variants replace the input (x, t) with

$$\phi_{\text{mem}}(x, t) = [x, t, t^\alpha, t^{1-\alpha}, t^{1-\beta}, \log(1+t), xt, xt^\alpha], \quad (4)$$

evaluated at $t_{\text{safe}} = \max(t, \epsilon)$ with $\epsilon = 10^{-8}$ and normalized by deterministic, analytically derived per-feature bounds rather than batch statistics, so that training and evaluation behavior coincide. The powers t^α , $t^{1-\alpha}$, $t^{1-\beta}$ are motivated by the regularity theory for time-fractional problems, which predicts t^α -type behavior near the origin [6]. The same feature builder is shared by both tracks, so no implementation asymmetry is introduced.

IV. EXPERIMENTAL PROTOCOL

All variants share (1), the residual construction, the collocation and boundary sampling, and the loss weights; only representation and architecture change. The core setting is a 50×50 discretization with mesh grading $r = 2$, $\alpha = \beta = 0.5$, and parameter counts matched within 11%. Every variant is trained for 8000 Adam epochs with cosine warmup over 500 epochs, learning rate 5×10^{-4} decaying to 10^{-5} , and each is run on the same ten seeds $\{0, 1, 2, 3, 4, 42, 123, 999, 2024, 2025\}$. The classical networks use three hidden layers of width 32; the surrogate uses a comparable embedding and mixing width chosen to match parameter count. Each step draws 32 interior collocation points, 16 boundary points and 16 initial points, with deterministic per-seed sampling so that two variants at the same seed see the same points. Boundary values are taken from the exact solution, since the problem is non-homogeneous. Errors are evaluated on a uniform 100×100 grid independent of the graded training mesh, over $t \in [10^{-10}, 1]$ to avoid the degenerate norm at $t = 0$, and the reported value is the error of the final model state, not the best observed during training. Runs are single-threaded on CPU, which makes the reported runtimes comparable across variants.

We report paired Wilcoxon signed-rank tests on final L_2 over the shared seeds, bootstrap 95% confidence intervals on the mean paired difference, and paired effect sizes. With

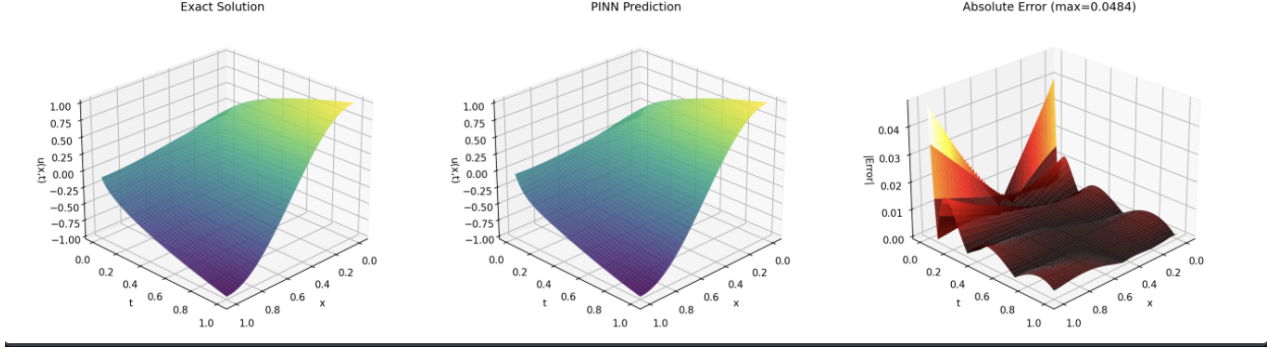


Fig. 1. Qualitative view of the manufactured problem: the exact field $t^\alpha \cos(\pi x)$, a representative classical PINN fit, and the absolute error surface. The error concentrates near $t = 0$, where the temporal derivative of the solution is singular, and near the spatial boundaries. This figure orients the reader on the problem and is not one of the benchmarked runs.

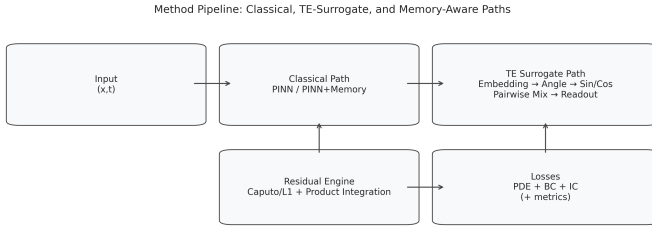


Fig. 2. Shared residual engine and the two model tracks. The Caputo/L1 and product-integration operators, the collocation sampling, and the loss weights are identical across variants; only the input representation and the network architecture change.

TABLE I
TEN-SEED RESULTS AT 8000 EPOCHS, $\alpha = \beta = 0.5$. RUNTIME IS SINGLE-THREAD CPU PER RUN.

Variant	L_2 (mean $\pm$ std)	L_∞	Par.	s
TE + memory	0.00262 $\pm$ 0.00081	0.0075	2498	96
Classical + memory	0.00415 $\pm$ 0.00173	0.0107	2433	50
TE + LayerNorm	0.01759 $\pm$ 0.00479	0.0583	2338	106
TE fixed residual	0.01834 $\pm$ 0.00380	0.0606	2306	95
Classical	0.03595 $\pm$ 0.00740	0.0880	2241	44
Zero predictor	1.00000	1.0000	—	—

five comparisons a Bonferroni threshold is 0.01, and we state explicitly which comparisons clear it.

Zero-predictor reference. Predicting $u \equiv 0$ gives relative $L_2 = 1$ by definition, and $L_\infty = \max |u_{\text{exact}}| = 1$ here. Any variant at or above that line has not learned the solution, and differences among such variants are not meaningful. This row is included in the results table.

V. RESULTS

A. Feature-Level Versus Architecture-Level Gains

Table I gives the primary result. Every variant now sits one to two orders of magnitude below the zero-predictor line, so the ordering reflects solution quality rather than residual noise. Because the residual engine, collocation sampling and loss weights are common to all variants (Fig. 2), the differences below are attributable to representation and architecture alone.

TABLE II
PAIRED TESTS ON FINAL L_2 , TEN SHARED SEEDS. THE BONFERRONI THRESHOLD FOR FIVE COMPARISONS IS 0.01.

Comparison (A vs B)	Ratio	Wins	p	d
Classical+mem vs Classical	8.66 $\times$	10/10	0.0020	−4.44
TE+mem vs TE fixed	6.99 $\times$	10/10	0.0020	−4.41
TE fixed vs Classical	1.96 $\times$	10/10	0.0020	−1.87
TE+mem vs Classical+mem	1.58 $\times$	9/10	0.0488	−0.80
TE+LayerNorm vs TE fixed	1.04 $\times$	7/10	0.4922	−0.18

Table II gives the paired comparisons. The largest single effect is the input representation: the classical-plus-memory control improves on the classical baseline by 8.66 $\times$ on every seed, with $p = 0.0020$ and $d = -4.44$. The same feature set produces a comparable 6.99 $\times$ improvement on the surrogate track. Both clear the Bonferroni threshold comfortably. Because the control differs from the baseline in the input alone, this improvement is attributable to representation and not to architecture.

The surrogate architecture also contributes independently of features: with identical (x, t) inputs, it improves on the classical baseline by 1.96 $\times$ on every seed, again at $p = 0.0020$. This is a genuine architectural effect, but it is roughly a quarter the size of the feature effect.

The combined model is strongest overall. Its margin over the representation-matched control is 1.58 $\times$, winning 9 of 10 seeds at $p = 0.0488$. We do not treat this as established: it does not clear the Bonferroni threshold of 0.01, and the bootstrap interval, while excluding zero, is narrow relative to the effect. The combined model is also the more expensive of the two, at roughly twice the per-epoch cost. We therefore report it as the leading variant rather than as a demonstrated win.

Worst-case error behaves consistently with the mean. The L_∞ column of Table I orders the variants identically to L_2 and spans roughly an order of magnitude across them, so the ranking is not an artifact of averaging over the evaluation grid. This matters here because the error field is not uniform: as Fig. 1 shows, it concentrates near $t = 0$ and at the spatial boundaries, so a method could in principle reduce mean error while leaving the worst-case behavior unchanged. That does not occur for any variant tested.

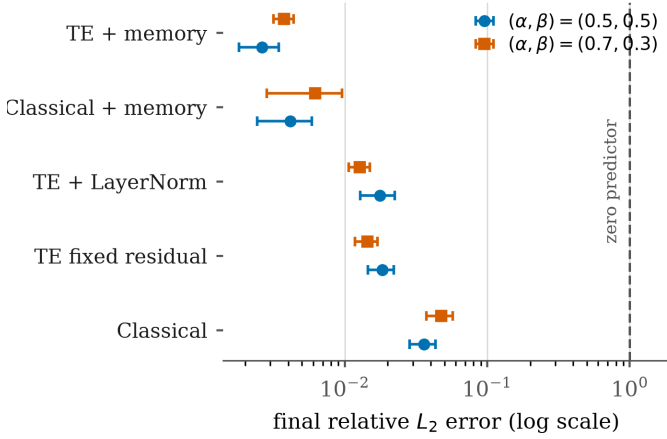


Fig. 3. Final relative L_2 error, mean $\pm$ standard deviation over ten seeds, at both fractional settings. Memory features move a variant by roughly an order of magnitude; the surrogate architecture contributes a smaller additional gain. The ordering is preserved at the altered setting.

B. Stabilization

The normalization variant is not supported. Against the plain surrogate it gives $1.04\times$, wins 7 of 10 seeds, and reaches $p = 0.4922$, with a bootstrap interval spanning zero. At a converged budget it is indistinguishable from the variant it was intended to stabilize.

C. Robustness

Repeating the full ten-seed protocol at $(\alpha, \beta) = (0.7, 0.3)$ preserves the ordering, as Fig. 3 shows alongside the primary setting. Memory features again dominate, at $7.67\times$ over the classical baseline on all ten seeds with $p = 0.0020$, and the surrogate architecture again contributes independently, at $3.29\times$ on all ten seeds with $p = 0.0020$. The margin of the combined model over the representation-matched control is stronger here than at the primary setting, at $1.64\times$ on 9 of 10 seeds with $p = 0.0098$, clearing the threshold that it missed above.

D. Sensitivity to Training Budget

The conclusions above depend on training to convergence, and we record this because it changes them. Under a severely truncated budget of 12 epochs all five variants score at or above the zero-predictor line, with final L_2 between 0.75 and 0.96 and four of five exceeding it in L_∞ , and the ordering inverts: the surrogate appears worst of the group rather than second. Rankings obtained in that regime measure initialization noise rather than model quality. This is a concrete argument for reporting the zero-predictor reference, which makes such a regime immediately visible.

VI. LIMITATIONS

The memory feature set contains t^α , which is exactly the temporal factor of the manufactured solution. A linear least-squares fit on the eight features alone reaches a relative L_2 of 0.124, against 0.362 for (x, t) , so roughly a factor of three

of the measured advantage is available before any network is trained. The trained advantage is $8.66\times$ and is therefore not explained by this alone, but the benchmark is favorable to the feature set, and the effect should be expected to shrink on problems whose time dependence is not t^α . We regard including t^α as principled, since α is a known coefficient of (1) and the regularity theory predicts this behavior, but the caveat belongs with the number.

Results are for one problem family, one spatial discretization, and two (α, β) settings. Ten seeds bound the achievable Wilcoxon p from below at 0.002, which is adequate for the large effects but marginal for the comparison between the combined model and the representation-matched control. All comparisons are equal-epoch; on an equal-runtime basis the surrogate variants are penalized by roughly a factor of two.

Finally, the architectural gain reported here is specific to the surrogate as described. Because the model is a learned Fourier-feature network, its advantage over a plain multilayer perceptron may be reproducible by other sinusoidal encodings that carry no quantum motivation at all; we have not separated those possibilities, and doing so would sharpen the attribution further.

Reproducibility. All variants are driven from declarative configuration files that fix the discretization, the architecture, the optimizer schedule and the seed list, so each reported run is reproducible from its configuration alone. The per-seed results underlying both tables and the figure are released as machine-readable records rather than only as summary statistics, which allows the paired tests to be recomputed independently.

VII. CONCLUSION

Under a fairness-controlled ten-seed protocol at a converged budget, analytic memory-aware input features are the dominant source of accuracy improvement for PINNs on a time-fractional integro-differential problem, exceeding any architectural change tested here by roughly a factor of four. A quantum-inspired trainable-embedding surrogate contributes a smaller but real and independent gain under matched inputs, and the combined model is strongest overall, although its margin over the representation-matched control is not established at the primary operating point. The normalization variant intended to stabilize the surrogate is not supported. We recommend that comparisons of this kind carry both a representation-matched control and an explicit zero-predictor reference. Without the first, feature-engineering gains are readily misattributed to architecture; without the second, rankings can be reported from a regime in which no model has learned the solution.

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